

# Design and Field-Programmable Gate Array Realization of a Multirate Multisampling Algorithm for Improving Signal-to-Noise Ratio in Pulse Compression Radars

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**Abstract**—Due to the ongoing advancements in small unmanned systems (SUSs), the field of study on detecting targets with small radar cross section (RCS) areas is constantly expanding. Due to their widespread use in both military and civilian fields, drones are considered the most significant class of small unmanned devices, garnering significant attention. As a result, numerous methods have been developed to improve radar detection performance by mainly increasing its processing gain (PG) in order to keep up with the advancement of drone capabilities. In this article, we introduce a multirate algorithm for improving the PG of the pulsed radar to enhance its detection performance of small RCS targets. The proposed method depends on acquiring multiple samples per subpulse from the received phase-coded signal and two coherent pulse intervals (CPIs) for decision-making. The simulation results are represented to show the provided PG to the system. Moreover, the field-programmable gate array (FPGA) implementation results and the utilized resources of the suggested algorithm are shown to demonstrate the superiority of our technique compared to other conventional methods.

**Index Terms**—Field-programmable gate array (FPGA) implementation, multirate signal processing, multisample per subpulse, pulse compression (PC) radars, radar cross section (RCS), signal-to-noise ratio (SNR) improvement.

## I. INTRODUCTION

THE last two decades have witnessed remarkable progress in the field of small unmanned systems (SUSs) because of the several advantages they can provide. SUSs, such as autonomous aerial vehicles, autonomous ground vehicles, and autonomous underwater vehicles, are robotic systems designed to operate without a human operator on board. These systems are typically compact and lightweight, and they are often equipped with sensors and other hardware to perform specific tasks [1]. Such systems can provide many benefits, as they can be quickly deployed and operated in areas that are difficult or dangerous for humans to access, such as disaster

zones, hazardous environments, and military battlefields [2]. Conversely, SUSs can pose a serious threat because they can be employed for lethal missions, including smuggling, espionage, and carrying bombs [3]. Because of their diminutive size and advanced materials, they often exhibit a low radar cross section (RCS) making them difficult to detect using conventional radar processing techniques. This limitation poses a significant challenge for existing radar technologies, as the traditional detection methods struggle to reliably identify and track such small targets. Therefore, in order to identify them, radar developers are compelled to search for advantageous ways to improve radar detection performance [4].

Enhancing the radar performance to detect small RCS targets as drones is basically dependent on raising the transmitted peak power, extending the transmitted pulse duration, or lengthening the integration time. These techniques are the main traditional strategies used to increase the pulsed radar signal-to-noise ratio (SNR). The first approach is significantly expensive and increases system power consumption, and the second approach increases the radar blind range in monostatic systems; it also degrades the radar range resolution [5]. Lengthening the integration time or increasing the number of samples captured per subpulse will boost the system PG. However, the computation time will significantly increase. Also, the required hardware resources will massively rise. Consequently, the field-programmable gate arrays (FPGAs), or the implementation boards, may be exceedingly costly or unavailable [6].

Because the aforementioned methods are considered conventional approaches, researchers are seeking effective methods to improve the radar SNR for small RCS targets' detection. Gong et al. [7] proposed the Doppler signal-to-clutter ratio (DSCR) detector instead of the SNR detector for improving system SNR to detect weak and small RCS targets accordingly. Since the DSCR method utilizes a lower threshold than a typical SNR detector, a large rise in the false alarm rate noticeably occurs. In addition, the authors' suggested method has not been implemented using a real-time processor, which does not give sufficient validation for the method to be applied in practical applications. In [8], a method for enhancing bistatic radar SNR is proposed. The technique methodology is based on Doppler and range compensation. It works by dividing the recorded data into multiple frames, and then,

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the fast Fourier transform (FFT) algorithm is employed to calculate the Doppler frequencies, which are used in the compensation process. The Doppler frequency compensation operation is accomplished by collecting 24 frames for the target and specifying the target Doppler frequency at each frame, and then, a linear regression tool is used to find the best line that fits the relation between frames (time) and the Doppler value. Therefore, the target Doppler frequency in the next frames can be estimated, and it can be compensated accordingly. The proposed method is recursive making it not suitable for real-time processing. Additionally, the SNR enhancement is limited to a maximum of 4.4 dB, with no possibility for further improvement.

Shen et al. [9] introduced an improved coherent integration technique through 2-D frequency correction for high-resolution wideband radars. The method operation is based on the realignment of the received data in a fast-time frequency-Doppler domain for compensating the target motion and achieving energy accumulation. This method has an obvious effect on high-resolution radars in which the across-range bin problem occurs in the case of dealing with high-speed targets. Moreover, this method will not provide a significant improvement to the performance of narrowband or moderate-band radars. The method performance was assessed through simulation and real-data processing; however, no details are provided about the practical realization or its applicability to other radar systems.

Jawad et al. [10] provided an algorithm for improving SNR for weak targets in passive radars based on receiver channelization. Its operation is based on determining the gap between the required SNR and the acquired SNR, and then, an iterative algorithm is used to determine the optimal number of channels for obtaining the required SNR difference. The channelization is done by dividing the receiver's bandwidth (BW) into overlapped  $N$  channels to reduce the received signal noise. The suggested method is not appropriate for real-time applications, as it depends on an iterative algorithm to determine the ideal number of channels to obtain the desired enhanced SNR value. Moreover, the proposed method's practical implementation results have not been discussed.

Another approach for improving the radar detection performance was investigated in [11]. The introduced method depends on exploiting the received signal from neighboring communication systems to enhance radar detection. The coexistence of the radar and communications system is demonstrated by the generalized likelihood ratio test (GLRT), which indicates better radar performance. This technique, however, only works well when the received communication signal has a high SNR. It also relies on nearby communication systems' assistance rather than a particular methodology to improve performance. The method's provided SNR enhancement is only around 3 dB. Moreover, its validation has depended only on numerical examples without any evidence to show if it is suitable for practical applications.

Youssef et al. [12] proposed a system that is based on low-rate coded time compression overlap-add technique. The system is based on transmitting various types of radar waveforms at the same time to take advantage of each waveform type. The proposed system increases the SNR by the same

factor as the increase in the sampling rate compared to the classical systems. As a result, it is complex to practically implement as the system has a separate processing channel for each waveform. Therefore, the utilized resources will be significantly high.

Wu et al. [13] introduced an algorithm for airborne maritime surveillance radar to improve the detection of moving targets. The airborne maritime surveillance radar is a radar system that is mounted on an aircraft for detecting and tracking targets on the sea surface. The proposed technique aims enhancement of targets' detection in sea environment by compensating the radar platform compensation and then realignment of the received range envelope to solve range migration issues. The method's validation depended only on simulation analysis without demonstrating the technique applicability for practical implementation. Furthermore, there are no simulation results introduced about the algorithm detection probability at different SNRs, which makes the algorithm evaluation incomplete. Also, in [14], a proposed algorithm had been introduced to enhance the SNR and the range resolution of frequency-modulated continuous-wave radars. The method relies on concatenating short chirps to increase the signal length for improving the system resolution. The radar SNR is enhanced by combining multiple chirps. However, the authors had not declared the improved PG value explicitly. In addition, the algorithm detection performance had not been evaluated under interference condition at different SNRs.

Although the previously discussed SNR improvement techniques have shown how to improve radar detection, some of them are iterative and unsuitable for real-time applications such as radar systems, and others mainly rely on simulation evidence without discussing their practical applicability. Furthermore, the provided PG improvement value is limited. In this article, our proposed multirate multisampling (MRMS) SNR improvement technique is introduced for enhancing radar detection for small RCS targets. The proposed method relies on acquiring multiple samples from each range cell in the received signal in order to increase the system PG and the detection capability accordingly. The MRMS algorithm relies on at least two coherent pulse intervals (CPIs) for making decisions. The suggested technique is designed to be practically implemented. Therefore, the FPGA implementation results of the MRMS algorithm are introduced along with the utilized implementation resources. The implementation outcomes have revealed the efficiency and superiority of the MRMS technique over other classical methods in terms of the used resources and the provided PG. Moreover, the proposed technique performance has been assessed by conducting a practical setup to acquire real-world data. The outcomes have proven the solid concept of the MRMS technique when working in a real environment. This article's findings can be summarized as follows.

- 1) The obtained PG from the proposed MRMS method can reach 8.1 dB, which can be increased by increasing the acquired samples per subpulse, consequently.
- 2) This article presents a detailed FPGA-based implementation of the proposed method with results to prove its practical feasibility.
- 3) The MRMS technique is not iterative; it operates on the real-data flow without using feedback or any recursion

algorithms, which emphasizes its suitability for real-time applications.

- 4) The suggested technique has been tested under hard conditions, such as Weibull-distributed clutter, and the results have ensured the efficiency of the MRMS method in improving the radar detection performance compared to other techniques that had not been tested under the same conditions.
- 5) The MRMS method has been evaluated through a practical setup under real-world conditions, demonstrating its robustness and superior performance. The results underscore its capability to operate effectively under diverse types of interference.

This article is organized as follows. Section II gives the MRMS algorithm's theoretical background. The simulation results are presented in Section III. Section IV gives the results of the FPGA realization and the hardware resources used. Section V includes a summary of this article and the future work.

## II. ALGORITHM METHODOLOGY BASIS

The methodology of the MRMS algorithm stands on determination of the desired range resolution  $\Delta R$  and the maximum range to be covered in order to specify the fast-time sampling frequency, and the number of samples  $N_r$  in the pulse repetition period (PRP). The aim of the proposed technique is to acquire  $n$  samples per range cell from the received phase-coded signal and add them together to enhance the system PG. The MRMS algorithm operation has to preserve the independence of received noise samples, which is assumed to be white noise of zero dependence. Therefore, the MRMS receiver has a BW of  $f_s/2$  Hz to avoid correlation between the sampled noise. Consequently, when the received samples are added together, the noise level will be minimized, and the system detection ability will be improved. The relation between  $f_s$  and the subpulse BW can be expressed as

$$f_s = n \times B_b = \frac{1}{T_s} \quad (1)$$

$$B_b = \frac{1}{\tau_b} \quad (2)$$

where  $B_b$ ,  $\tau_b$ , and  $T_s$  are the subpulse BW, the subpulse temporal length, and the sampling time, respectively. The utilized pulse compression (PC) waveform in our technique is a binary-phase-coded signal that has an autocorrelation function (ACF) with the best peak-to-sidelobe level (BPSL). This type of waveform was fully introduced in [15]. The transmitted waveform  $X_t$  can be expressed as

$$X_t(t) = \begin{cases} \sum_{k=0}^{N_p-1} a_t e^{j\phi_k} (q(t - k\tau_b)) & 0 \leq t < N_p \tau_b \\ 0, & N_p \tau_b \leq t < N_r - 1 \end{cases} \quad (3)$$

$$q(t) = (u(t) - u(t - \tau_b)) \quad (4)$$

where  $q(t)$  is a pure subpulse with a time duration  $\tau_b$ .

Here,  $a_t$  is the transmitted amplitude,  $\phi_k$  is the phase of each subpulse switching between 0 and  $\pi$  according to the used binary code,  $N_p$  is the biphasic code length, and  $k$  is the

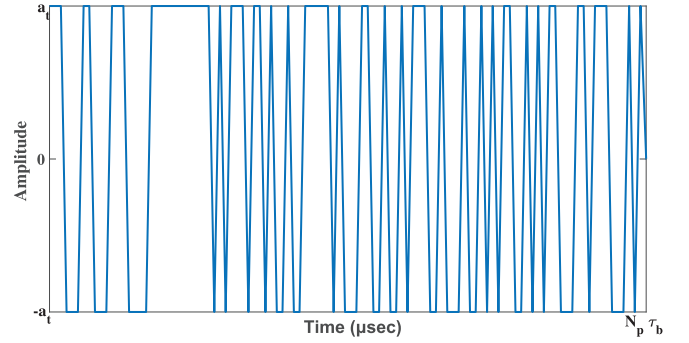


Fig. 1. BPSL biphasic modulated signal shape.

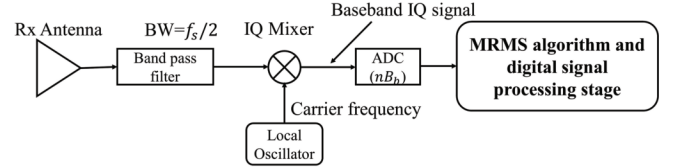


Fig. 2. Block diagram of the radar system receiver using the MRMS technique.

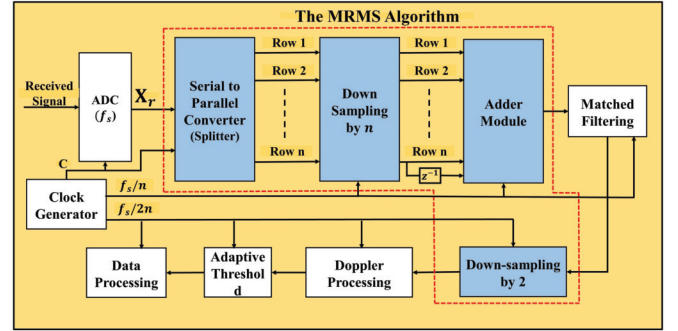


Fig. 3. Block diagram of the proposed MRMS technique.

subpulse index. Fig. 1 shows the ideal signal shape without considering any shaping techniques.

Fig. 2 displays the block diagram of the radar system receiver of the proposed MRMS technique. The system BW is  $f_s/2$ , and the in-phase and quadrature mixer and the local oscillator manage the downconversion process. At the analog-to-digital (ADC) module, the fast-time sampling frequency  $f_s$  samples the resulting baseband complex signal  $X_r$ , producing  $nN_r$  samples in a single PRP.

The MRMS receiver is designed to have a higher sampling frequency and, consequently, a broader BW than the classical radar receiver, which allows for acquiring much more of received signal energy. Therefore, the system sensitivity will not be deteriorated. Also, the proposed MRMS method depends on the addition of the received independent samples, which causes a decrease in the received noise amplitude. The  $n$ -sampled signal at the ADC output can be represented as

$$X_r = [n_{11} \ n_{12}, \dots, n_{1n}, \dots, s_{l1} \ s_{l2}, \dots, s_{ln}, \ s_{(l+1)1} \ s_{(l+1)2}, \dots, s_{(l+1)n}, \dots, s_{(l+N_p-1)1} \ s_{(l+N_p-1)2}, \dots, s_{(l+N_p-1)n}, \dots, n_{N_r1} \ n_{N_r2}, \dots, n_{N_rn}] \quad (5)$$

where the  $s_{xy}$  terms refer to signal plus interference samples and the  $n_{xy}$  terms refer to only interference samples. The suffix  $xy$  is the subpulse (range resolution cell) number and

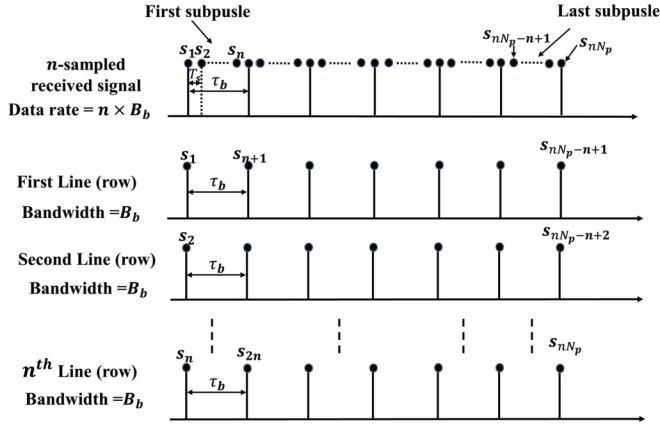


Fig. 4.  $n$  output parallel lines after the STP and downsampling by  $n$  processes when the target range is optimal.

the sample order within the subpulse.  $l$  is the range bin in which the target is received.

Fig. 3 represents the block diagram of the MRMS algorithm. The resulting ADC output data will flow serially to a splitter unit represented by the serial-to-parallel (STP) module. Because the aim of the proposed algorithm is to add the  $n$  samples from every range bin together, an STP module is employed to split the serial data into  $n$  parallel data lines for the addition process. As shown in Fig. 4, the first line is the  $n$ -sampled received data. The STP distributes every  $n$  sample of each range cell over its  $n$  output lines. Consequently, the first output line contains the first sample from the range cell, and the  $n$ th line contains the  $n$ th sample from the same range cell. In other words, the  $n$ -sampled pulse will be split into  $n$  pulses; each pulse is contained in a line from the STP module output lines. The addition of the split samples will enhance the target signal component and lessen the noise amplitude since the obtained samples are independent of uncorrelated noise. This will have a significant positive effect on the target detectability process. The data lines from the STP can be mathematically expressed as a data matrix, as shown in the following equation:

$$\begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} n_{11} & \cdots & s_{l1} & \cdots & s_{(N_p+l-1)1} & \cdots & n_{N_r1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ Y_n & n_{1n} & \cdots & s_{ln} & \cdots & s_{(N_p+l-1)n} & \cdots & n_{N_rn} \end{bmatrix} \quad (6)$$

where  $Y_1, Y_2, \dots, Y_n$  refers to the  $n$  parallel data rows resulted from the STP module. Every column in the data matrix represents the  $n$  samples taken from a certain range gate, and every row from the  $n$  data rows represents a part of the received signal. After the STP module, the  $n$  rows are then directed to a downsampling by the  $n$  module. The output is also  $n$  parallel lines triggered by  $1/n$  of the ADC clock  $f_s$  resulting in a subpulse with a BW  $B_b = 1/\tau_b$ . Fig. 4 shows the operation of these two modules.

The  $n$  lines resulting from the downsampling by the  $n$  module will be added to obtain the desired PG. The resulting signal sequence,  $Y_s$ , after addition is written as follows:

$$Y_s = N_1, N_2, \dots, S_l, \dots, S_{N_p+l-1}, \dots, N_{N_r} \quad (7)$$

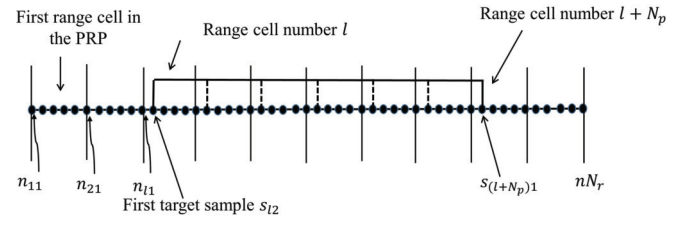


Fig. 5.  $n$ -sampled target arrived at a sample whose number is not an integer multiple of  $n$ .

where

$$Y_s = \sum_{i=1}^n Y_i, \quad N_1 = \sum_{i=1}^n n_{1i}, \quad S_1 = \sum_{i=1}^n s_{li} \\ S_{N_p+l-1} = \sum_{i=1}^n s_{(N_p+l-1)i}, \quad N_{N_r} = \sum_{i=1}^n n_{N_r i} \quad (8)$$

Each summation symbol represents the addition of the  $n$  samples from a certain range bin. The  $N$  terms are added noise samples, and the  $S$  terms are the target signal-added samples. The amplitude of the added noise samples has decreased due to the destructive summation and the statistical independence between them, while the amplitude of the added target samples has increased, resulting in a significant increase in the PG.

The received sampled signal expressed in (5) is the most optimal case to happen because the target has arrived at the start of a certain range gate. In this case, as shown in Fig. 4, the target arrival sample  $s_1$  must be an integer multiple of  $n$ , which causes the rows of the data matrix to be aligned together, such that the first sample in each range gate comes at the top of the matrix column, and the last sample in the subpulse is the last sample in the same column. In this case, when the  $n$  lines (rows) are summed in the adder module, the energy of the target will be enhanced significantly.

Generally, the target range is unknown. Therefore, it is not guaranteed to receive the returned target pulse at the start of the range bin, as described in Fig. 5. The figure shows the case of an  $n$ -sampled received signal from a target at a certain sample within the range cell number  $l$ . In this case, the arrival sample is not an integer multiple of  $n$ .

In this case, the received signal  $X_R$  at the ADC output can be expressed as

$$X_r = [n_{11} \ n_{12}, \dots, n_{1n}, \dots, n_{l1} \ s_{l2} \ s_{l3}, \dots, s_{ln}, s_{(l+1)1} \\ s_{(l+1)2}, \dots, s_{(l+1)n}, \dots, s_{(l+N_p-1)1} \ s_{(l+N_p-1)2}, \dots, \\ s_{(l+N_p-1)n}, \ s_{(l+N_p)1} \ n_{(l+N_p)1}, \dots, n_{(l+N_p)n} \\ n_{N_r1} \ n_{N_r2}, \dots, n_{N_rn}]. \quad (9)$$

After the STP process, the data row output will be expressed as

$$\begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix} = \begin{bmatrix} n_{11} & \cdots & n_{l1} & \cdots & s_{(l+N_p-1)1} & s_{(l+N_p)1} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ n_{12} & \vdots & s_{l2} & \vdots & s_{(l+N_p-1)2} & n_{(l+N_p)2} & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ n_{13} & \vdots & \vdots & \vdots & s_{(l+N_p-1)3} & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \cdots & \vdots & \cdots & \vdots & \cdots & \vdots \\ X'_n & n_{1n} & \cdots & s_{ln} & \cdots & s_{(l+N_p-1)n} & n_{(l+N_p)n} \end{bmatrix} \quad (10)$$

where  $X'_n$  is a vector of length  $N_r$  samples. In the case described by (10), it is noticed that the first sample of target

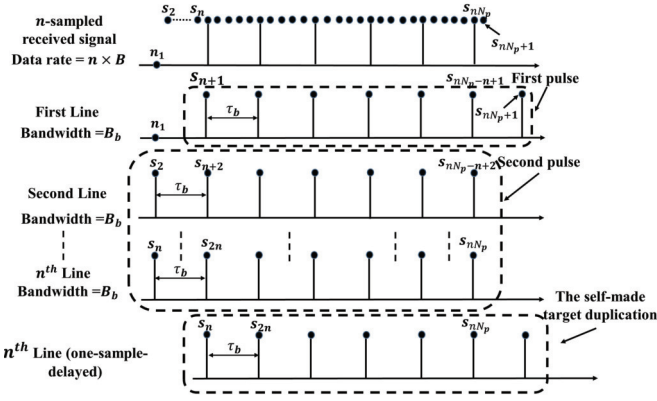


Fig. 6. Sample number at which the target arrived is not an integer multiple of  $n$ , and the target duplication scenario will occur.

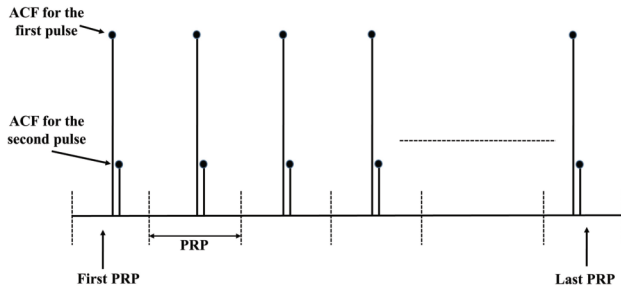
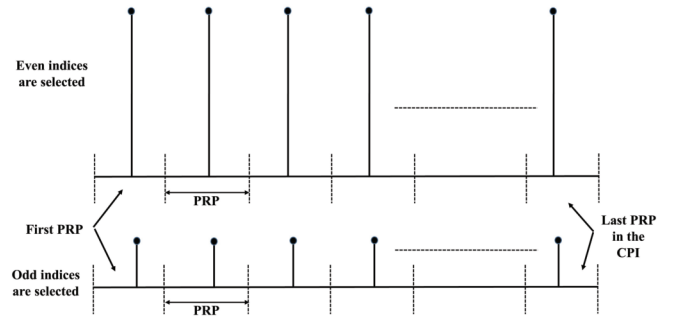


Fig. 7. MF output for a single target after the self-made target duplication.

return is  $s_{l2}$  that is not the first sample in its corresponding subpulse (column), but it is shifted by one sample as the target is delayed by one sample from the start of the range cell. As a result, the obtained  $n$  parallel lines represented in the matrix rows are not aligned together. When these lines are added together in the adder module, they will represent two pulses superimposed on each other; one is delayed from the other by one sample. The first pulse has an amplitude of  $n-1$  multiples of the original received pulse due to addition, and the second pulse has an amplitude equal to the original received signal amplitude. In general, the two pulses' amplitudes vary according to the order of the target arrival sample within the  $n$  samples of the range cell. When the first sample comes at the beginning of the range gate, the target will be represented by one pulse, but if it comes at any other sample, the splitting process will result in two pulses from the same target. We refer to this case as the "Target Duplication" case.

To solve this issue, we have to unify the reception scenarios by making all targets always be represented by two pulses. This can be achieved if a one-sample-delayed version of the  $n$ th data line is added with the original  $n$  lines, as shown in Fig. 6. In this way, every target return is forced to be represented by two pulses. In addition, the corresponding ACF will also be composed of two samples. Using this method, we have unified the MF output for all targets, regardless of their ranges.

Fig. 7 shows the MF output for the target return after the self-duplication that is made by adding the delayed version of the  $n$ th data line. Every target will be represented by an ACF composed of two samples representing the two pulses of the target. The ratio between the two ACFs depends on the



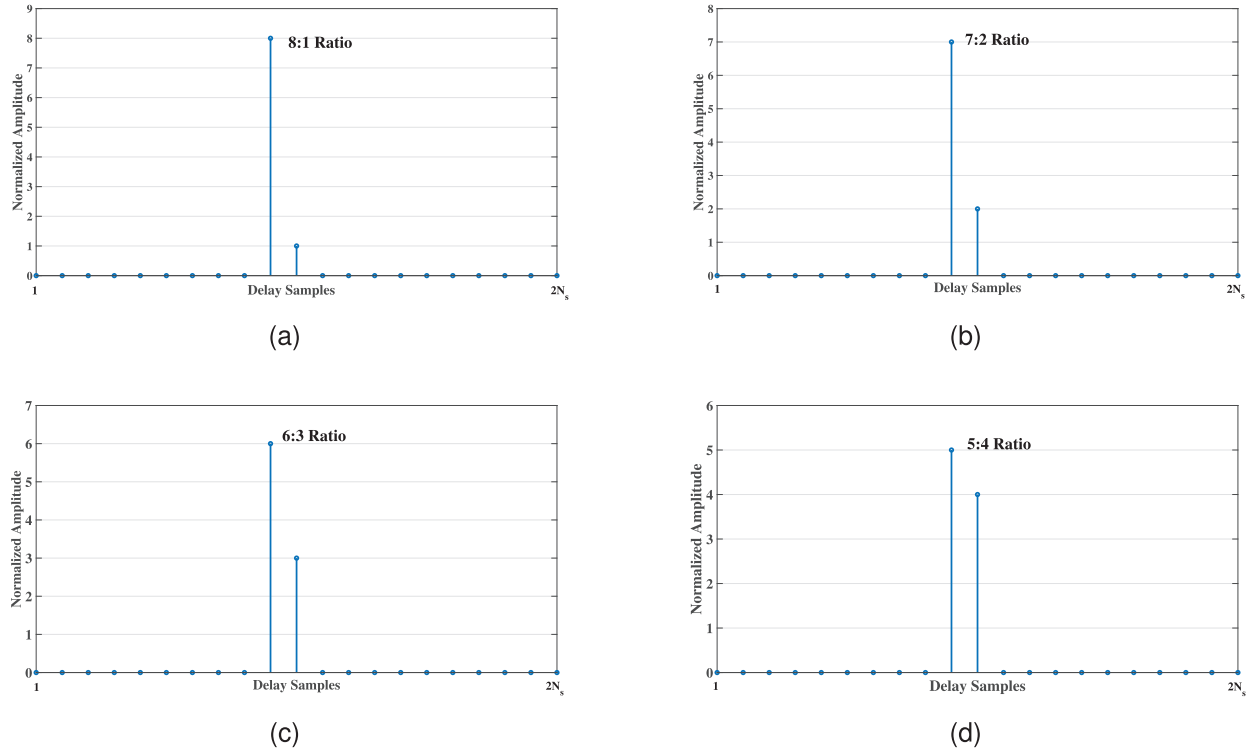


Fig. 9. Expected four ratios between the two ACFs representing one target. (a) 8:1 ratio. (b) 7:2 ratio. (c) 6:3 ratio. (d) 5:4 ratio.

the proposed technique. The parameters used in the simulation are listed in Table I.

The operating waveform is a binary phase-coded signal that has an optimal peak-to-sidelobe ratio, as reported in [15]. Regarding the desired coverage range and the design range resolution  $\Delta R/2$ , the number of samples in one PRP is 1024. At the receiver, the target echo is sampled at a rate eight times the subpulse BW  $B$  to increase the PG. Using the formula in (11) with  $n = 8$ , the obtained ACF ratios are 8:1, 7:2, 6:3, 5:4, 4:5, 3:6, 2:7, and 1:8. It is noticed that each ratio has an identical opposite. For example, the 8:1 ratio is identical to the 1:8 ratio, but the order of the samples is reversed. If the enhanced sample appears at an even index in the 8:1 ratio, it will correspond to an odd index in the 1:8 ratio. As the algorithm relies on using the odd sample for one CPI and the even samples for the successive CPI, there are  $n/2$  ratios for even  $n$ , and for the odd  $n$ , there are  $(n + 1)/2$  ratios.

To declare the effectiveness of our algorithm, the eight samples per range cell were taken as a case study. There are four ACF ratios, which are particularly 8:1, 7:2, 6:3, and 5:4. Fig. 9 shows the four expected ratios between the two ACFs' samples of the target at SNR = 0. Note that these results are obtained at the MF output before downsampling by two stages.

For evaluating the proposed algorithm, the probability of detection (PD) has to be investigated with and without the proposed algorithm to highlight the MRMS technique positive effect on the system performance. In Fig. 10, the PD of a conventional radar processor and the PD of the proposed MRMS algorithm are both plotted at different SNRs considering the noise type as white Gaussian noise. The figure also shows the obtained PG at each expected scenario from the four scenarios. As shown in Fig. 9, the ratios 8:1, 7:2, 6:3, and 5:4 have

a corresponding PG of 8.1, 7, 5.7, and 4.1 dB, respectively. From the obtained results, we can deduce that as the ACF ratio approaches one, the PG reaches its minimum limit of 4.1 dB gradually. Furthermore, only  $n/2$  situations are taken into consideration for evaluating the system; the ratios 1:8, 2:7, 3:6, and 4:5 have the same PG as the ratios 8:1, 7, 5.7, and 4.1 dB, respectively.

To further evaluate the MRMS algorithm, the method has been tested in a scenario involving Weibull-distributed clutter with a scale parameter of 2.5 and a shape parameter of 0.9. The clutter distribution parameters are selected to emulate the operating environment conditions. The results, depicted in Fig. 10, demonstrate that the MRMS algorithm consistently enhances the signal processor's processing gain (PG) effectively, even in the presence of clutter. The PGs achieved by applying the MRMS algorithm at various signal-to-clutter plus noise ratios (SCNRs) remain unchanged before and after the introduction of clutter, emphasizing the algorithm's adaptability under diverse conditions.

#### IV. FPGA IMPLEMENTATION RESULTS AND RESOURCE UTILIZATION

This section is divided into two sections. IV-A will present the implementation results at each stage of the suggested processor, and IV-B will show the utilized hardware resources of the MRMS technique compared with other classical methods.

##### A. Implementation Results

The proposed MRMS algorithm is designed using the ISE 14.7 Design Suite, and it is implemented on the XC7A200T-FBG-484 FPGA board. Implementation data at each stage

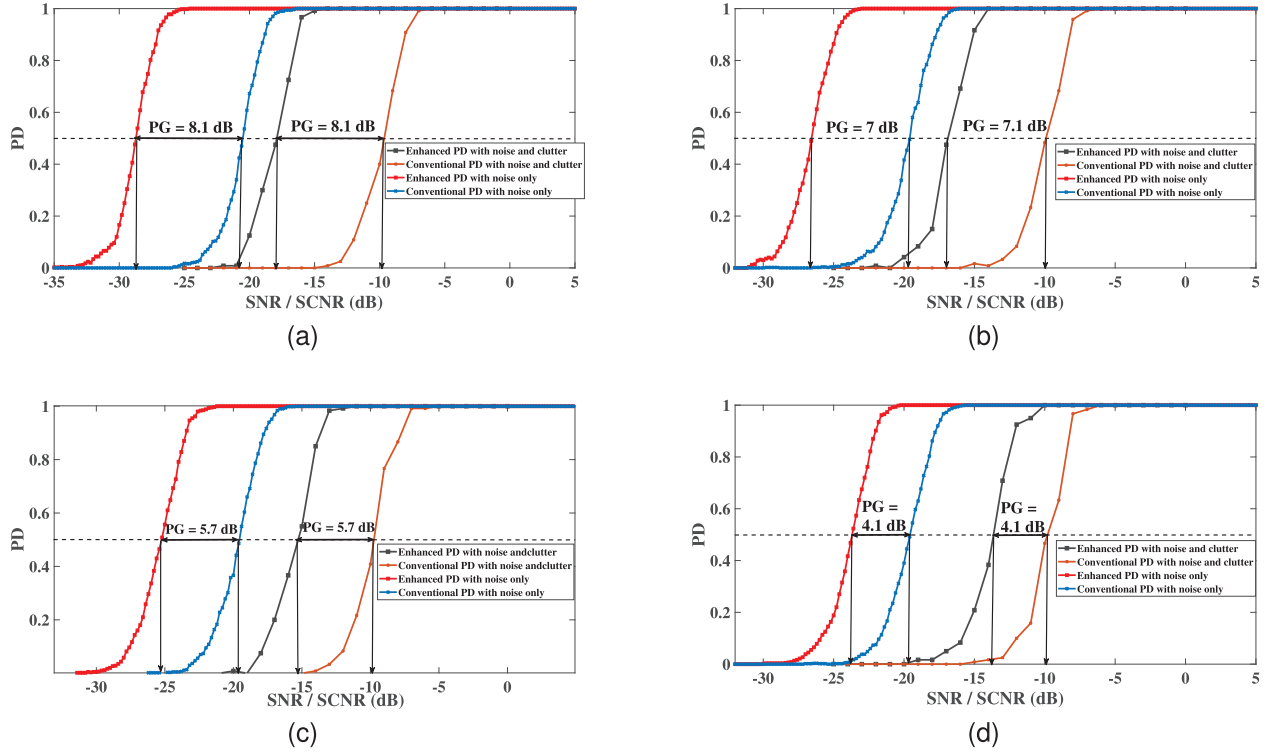


Fig. 10. Comparison between the conventional radar PD (blue and orange lines with cross markers) and the improved PD (red and black lines with square markers) after applying the proposed algorithm at different ACF ratios with and without Weibull-distributed clutter. (a) 8:1 ratio, (b) 7:2 ratio, (c) 6:3 ratio, and (d) 5:4 ratio.

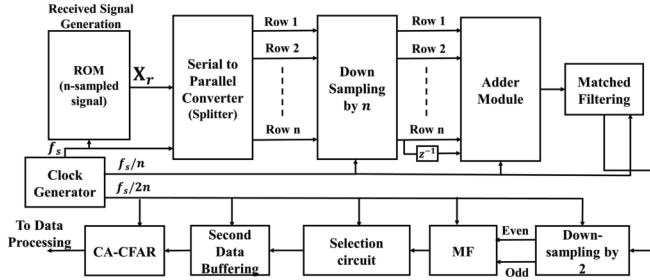


Fig. 11. Detailed implementation block diagram of the MRMS algorithm.

of our processor are captured by the ChipScope debug core. ChipScope is a set of tools provided by AMD to allow for easy probing of the internal FPGA signals [16]. The implementation setup is composed of a 12-V power supply to run the FPGA board. To simulate the ADC operation, the received waveform is digitally generated in the FPGA using a read-only memory (ROM) module containing the hexadecimal representation of the  $n$ -sampled baseband signal coefficients presented in the single-precision floating-point format. The generated signal is time-delayed to emulate the target delay time. The same processing sequence is used for evaluating the classical SNR enhancement processors. The main goal of the experiment is to evaluate the proposed processor and compare the method's resource consumption to other classical SNR improvement techniques under identical conditions.

Table II presents the used implementation parameters in the signal generator and processor design.

Fig. 11 illustrates the MRMS algorithm implementation block diagram in more detail. The whole design is based on

TABLE II  
MRMS ALGORITHM IMPLEMENTED PARAMETERS

| Parameter                           | Value           |
|-------------------------------------|-----------------|
| Desired range resolution $\Delta R$ | 20 m            |
| Pulse duration $\tau$               | 7 $\mu$ sec     |
| Subpulse duration $\tau_b$          | 33.34 nsec      |
| Sampling frequency $f_s$            | 120 MHz         |
| Pulse repetition period             | 34.13 $\mu$ sec |
| Number of pulses in a CPI ( $M$ )   | 64 pulses       |
| Number of samples per subpulse $n$  | 8 sample        |
| ACF ratio                           | 8 : 1           |
| Probability of false alarm          | $10^{-6}$       |
| Receiver BW                         | 60 MHz          |
| ADC part number                     | ADS5541         |
| ADC resolution                      | 14 bits         |
| ADC ENOB                            | 11.3 bits       |
| ADC SNR                             | 73.2 dBFS       |

the single-precision floating-point format with a word length of 32 bit, which allows for representing a wider range of integer values than the fixed point format [17]. At the receiver side, the backscattered signal is sampled by 120 MHz to acquire eight samples per range cell. Fig. 12 shows the real part of the received signal after being sampled by eight samples per subpulse.

After the ADC operation, the sampled data are split into eight lines, downsampled by 8, and added together with a one-sample delayed version of the eighth line. The resulted data from the adder chain have a BW of 15 MHz, and the number of samples in the PRP becomes 1024 samples. Fig. 13 shows the adder chain output.

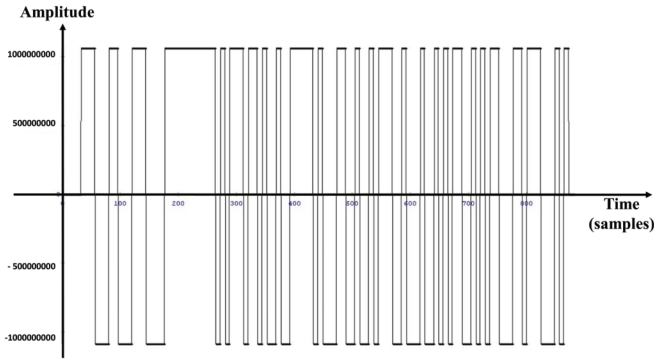


Fig. 12. Real part of received signal at the ADC output.

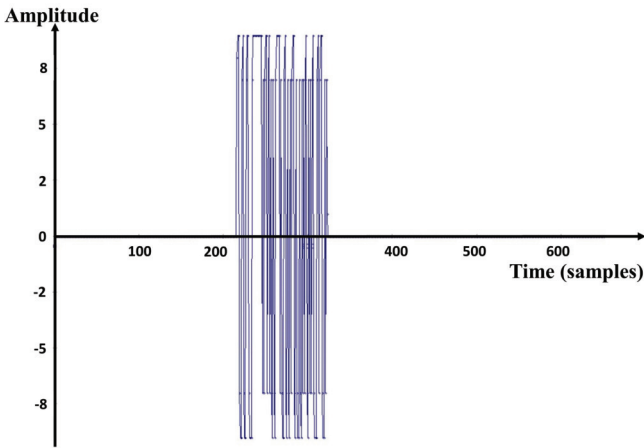


Fig. 13. Resulted signal at the adder chain output for the 8:1 ratio.

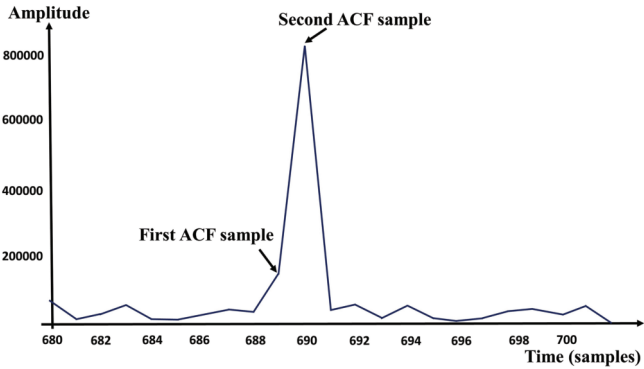


Fig. 14. ACF of a single target composed of two samples at SNR = 0 dB.

The signal is then applied to the MF, which is implemented in the frequency domain using the 1024-point FFT modules. The ROM units are used in the MF design to store the spectral coefficients of the reference signal. The received signal is transformed into the frequency domain by the FFT module, and the output is multiplied by the spectral reference signal coefficients using a complex multiplier; then, the inverse FFT is applied on the output data to retrieve the ACF of the signal in the time domain. The resulted ACF is composed of two samples with a ratio of 8:1, as shown in Fig. 14. The two samples are resulted because every target is represented by two pulses superimposed on each other.

After the MF process, the resulting data are downsampled by 2 to eliminate the target duplication in the MF output data. The output data from the downsampling by two processes are

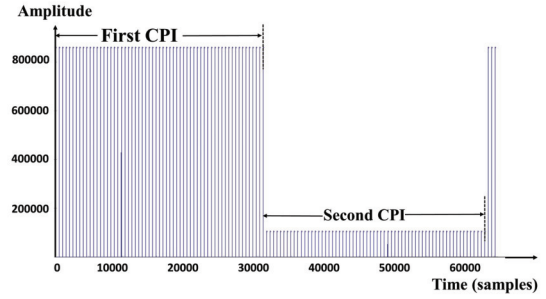


Fig. 15. Output of the selection module after the downsampling by two processes.

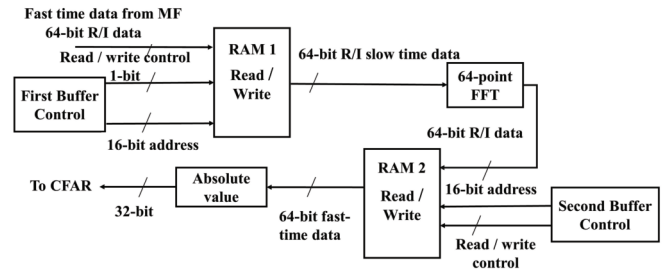


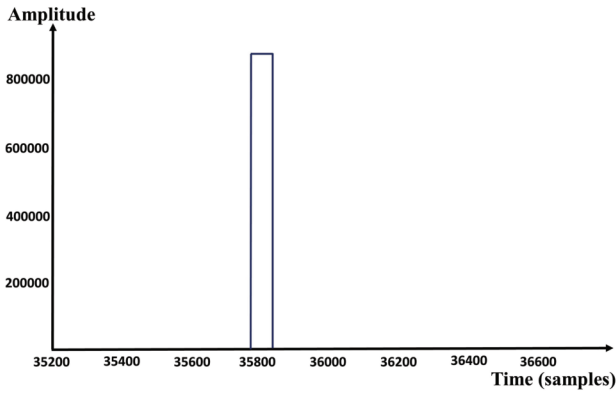
Fig. 16. Implementation block diagram of the data reordering and Doppler processing.

basically two data lines with a 7.5 MHz BW, and the number of samples per the single PRP becomes 512. These two lines are directed to the selection module that chooses one line for a CPI, and the second line is chosen in the next CPI. The number of samples in one CPI is  $64 \times 512 = 32768$  samples. The downsampled data at the output of the selection module are shown in Fig. 15.

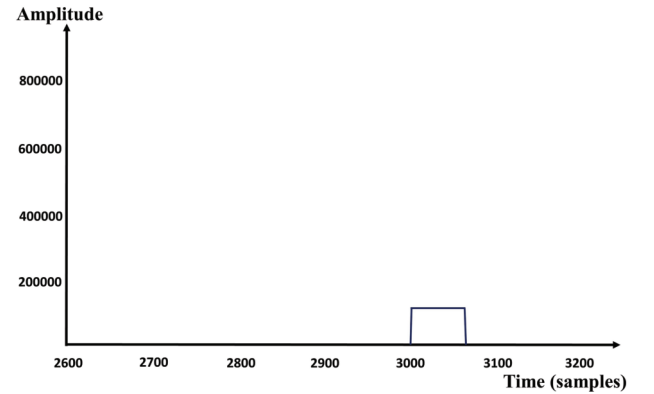
After the downsampling by two processes, the resulted fast-time data are reordered by a buffering circuit to formulate the slow-time data, which is required for the second integration to determine the target Doppler frequency. The buffering network is composed of two dual-port random access memory (RAM) modules and an FFT module. The first port of the first RAM receives the data in its original received order, while the second port is in the read mode. The address of the read operation is hopping by a step of 512 samples (samples in the PRP). The resulting data output becomes the required slow-time data. This operation ends at a time equal to the CPI, and the time calculation is accomplished by a 16-bit counter ( $512 \times 64 \times 2$ ). Once the CPI is completed, the operation is inverted, with the first port of the RAM switching to read mode and the second port to write mode. The whole addressing and control operation is manipulated by a circuit named buffer control, as shown in Fig. 16. The slow-time data formulated for a zero-Doppler target for the first and second ACF samples (even and odd) are shown in Fig. 17(a) and (b), respectively.

The buffered slow-time data are directed to the 64-point FFT to specify the target Doppler frequency. Fig. 18 shows the second FFT output; the peak index refers to the Doppler cell number, which is related to the target Doppler value.

The slow-time data at the output of the second FFT are reordered to get the fast-time (range) data in order to be applied to a cell-averaging constant false alarm rate



(a)



(b)

Fig. 17. (a) Data ordering of samples with even indexes for the Doppler processing. (b) Data ordering of samples with odd indexes for the Doppler processing.

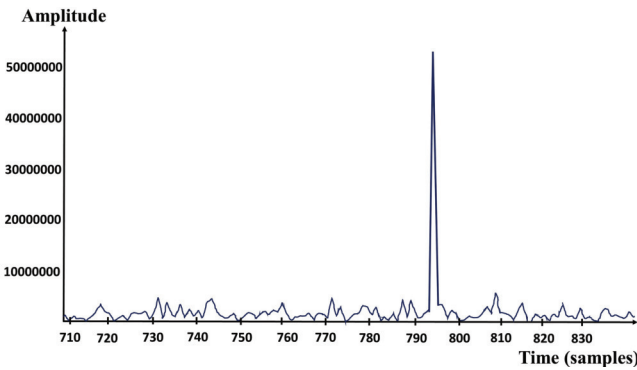


Fig. 18. Data magnitude at the second (Doppler) FFT module output.

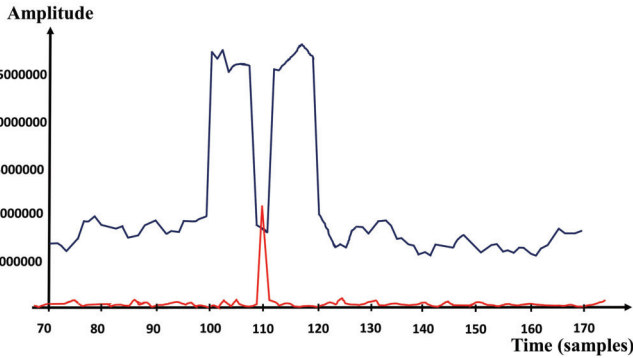


Fig. 19. CA-CFAR threshold is compared to the data output from the second integration.

(CA-CFAR) for target decision-making. The second buffer network is composed of true dual-port RAM and the second buffer control circuit. The second buffering circuit functions in the same way as in the first data buffering circuit, except that the hopping step is 64. The resulting complex fast-time data are applied to a CA-CFAR after getting its absolute value. The CA-CFAR is implemented with a window size of 12 cells and one guard cell. The CFAR parameters are selected to get a false alarm rate of  $10^{-6}$ . Fig. 19 shows the CFAR threshold compared to the absolute of the fast-time data.

### B. Resource Utilization Comparison

The supremacy of our suggested MRMS algorithm is demonstrated by comparing the FPGA device resource utilization when applying the MRMS approach and the other

TABLE III  
RESOURCE UTILIZATION COMPARISON BETWEEN THE MRMS  
ALGORITHM AND OTHER TECHNIQUES

| Method                                        | PG                       | BRAM | Slice<br>Reg-<br>is-<br>ters | Slice<br>LUTs | DSP<br>Slices | LUTs-<br>FF<br>Pairs |
|-----------------------------------------------|--------------------------|------|------------------------------|---------------|---------------|----------------------|
| Classical 2 samples<br>per subpulse           | 3<br>dB                  | 90%  | 17%                          | 28%           | 20%           | 42%                  |
| Classical 4 sam-<br>ples per subpulse         | 6<br>dB                  | 178% | 35%                          | 42%           | 44%           | 79%                  |
| Classical 8 samples<br>per subpulse           | 9<br>dB                  | 350% | 76%                          | 83%           | 95%           | 156%                 |
| MRMS algorithm<br>(8 samples per<br>subpulse) | 4.1<br>dB<br>at<br>least | 41%  | 18%                          | 31%           | 20%           | 38%                  |

conventional techniques under the same operation parameters. Table III shows the utilized resources for the proposed algorithm and other classical techniques for PG improvement. The classical method (two samples per subpulse) has almost complete utilization of the available block RAM (BRAM) count, while the MRMS algorithm uses less than half of its BRAM usage percentage. Both techniques are equal in the consumed DSP slices. The two samples per subpulse processor have insignificantly efficient utilization of slice registers, LUT-flip-flop pairs, and slice lookup tables (LUTs). However, the provided PG is only 3 dB, while the least attainable PG in the MRMS algorithm is 4.1 dB. Both the four samples and the eight samples per subpulse technique have exceeded the resource budget available in the utilized FPGA device. These techniques may provide significant PG, but they still cannot be synthesized on the target device due to resource shortages. Other techniques demonstrated in IV-A, such as the DSCR detector method in [7], the GLRT-based method in [11], and the BeiDou GEO satellite-based method in [8], did not reveal the practical processing steps and the utilized hardware as we have done in the MRMS algorithm.

### C. Proof of Concept

To validate the simulation and the FPGA implementation results, we have conducted a simple setup to evaluate the system performance with real data. The setup is composed of two software-defined radios (SDRs) for transmitting and

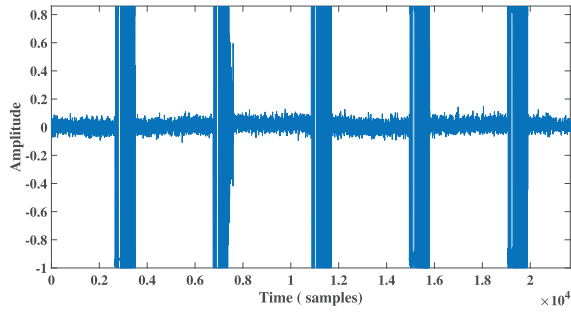


Fig. 20. Part of the received CPI.

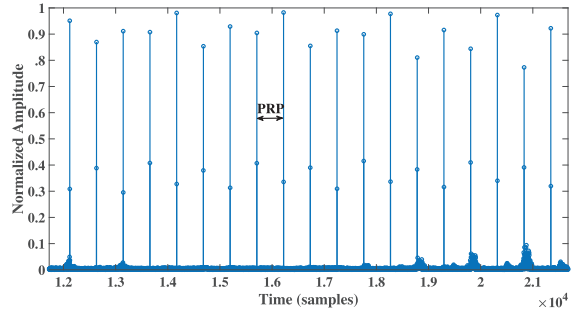


Fig. 21. ACFs of successive pulses of a single target are composed of two samples.

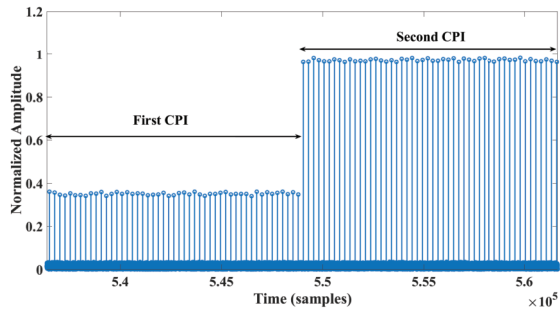


Fig. 22. Output of the selection module.

receiving the signal. The SDRs models are ADALM-PLUTO manufactured by analog devices [18]. The setup is operating in the L band at 2.4 GHz, with an appropriate distance between them. The transmitted signal from the transmitter SDR module has a 2.5-MHz BW. It emulates a backscattered signal from a small RCS target. The received signal is sampled at 20 MHz to acquire eight samples from each received range bin. The received signal is shown in Fig. 20.

The data, after being  $n$ -sampled, are applied to the STP module to separate the data into eight lines, and then, the output is added to enhance the system PG. The added output is applied to the matched filter. The ACF of a single target is shown in Fig. 21.

The MF output is downsampled by 2 in the selection circuit resulting in two output data lines, one for the odd indices and the other containing the even index data of the MF output, as shown in Fig. 22.

The resulting data from the selection module are applied to the Doppler processing to carry out the second integration and estimate the target Doppler frequency, as illustrated in Fig. 23.

From the presented results, we can ensure the efficiency of the proposed MRMS algorithm as its processing sequence

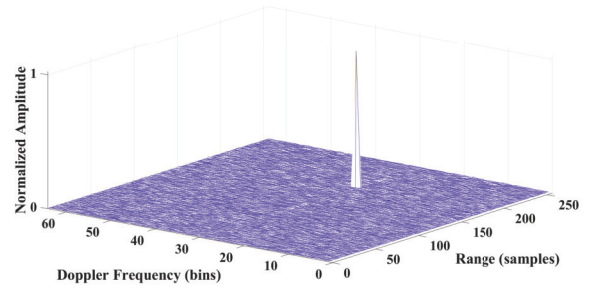


Fig. 23. Second integration (Doppler integration) output.

outcomes are similar to the simulation and the FPGA implementation results, which proves its practicality and efficiency.

#### D. Discussion

The main goal of the MRMS method is to cover a specific research gap, which is enhancing the signal processor PG without a proportional increase in the utilized resources and a real-time manner. The MRMS algorithm has fulfilled a 8.1-dB PG that can be increased according to the number of acquired samples per subpulse. Unlike other techniques, the MRMS method has managed to combine the feasibility for hardware implementation, suitability for real-time processing, and the compelling PG it can provide. Additionally, the MRMS algorithm was tested under harsh environmental conditions such as heavy-tailed clutter, and it has proven its superiority in enhancing the radar processor performance, maintaining the same PG for all reception scenarios.

The findings of the MRMS method can be very useful for radar manufacturers and researchers who aim to find practical ways that can be hardware-applied to boost the system PG because the MRMS methodology is to design a method to enhance the system's performance besides being feasible for practical implementation to serve real-time applications with moderate resources. Also, applying the MRMS algorithm on phase-coded signals can be considered as a clue to be applied on other types of waveforms for different purposes. The algorithm findings can be stated as follows.

- 1) The best attainable PG is 8.1 dB. Without applying our algorithm, a 8.1-dB processing could be obtained if the number of integrated pulses in the Doppler processing is increased by almost six times. However, this would necessitate a proportionally significant increase in the implementation resources, which might be impractical for the processor's capabilities. Additionally, the computation time will increase significantly.
- 2) The second approach to obtain a 8.1-dB PG is by widening the pulse duration by almost 6.5 $\times$ , which makes the pulse much more sensitive to Doppler frequency in the case of a phase-coded signal [19]. Also, the radar minimum detectable range will increase, and the hardware resources required for implementation will also increase.
- 3) The attainable PG from the proposed algorithm improves with increasing the samples taken from the range cell. This requires an ADC with a high sampling rate. Although it is expensive, it is available on the market.

Fortunately, recent ADCs can provide a sampling frequency of 6 GHz.

- 4) The MRMS algorithm performance under clutter condition is solid; it provides the same PG in all reception scenarios, which validates its superior performance and its operation adaptability under different conditions of interference.
- 5) The MRMS algorithm has been practically evaluated in real-world condition, and it has proven its effectiveness and supremacy, which ensures its full adaptability to operate in real environments.

## V. CONCLUSION

The article has presented the theoretical foundation through MATLAB simulation along with the FPGA implementation of our proposed MRMS algorithm. Additionally, the attainable PG of our method is represented for every scenario depending on the target range and the number of samples  $n$  taken from every range cell. As per the analyzed case study of eight samples per subpulse, applying the MRMS algorithm has improved the target SNR by at least 4.1 dB. In the presented case study and according to the certain FPGA board used for evaluating all the SNR improvement techniques in the comparison, namely, 59% of the BRAM resources of the board are still available. Therefore, the acquired samples per chip can be doubled, leading to a significant increase in the PG without any suffering from resource shortage. Additionally, the proposed algorithm was tested in a real-world scenario using the SDR setup. The results demonstrated the MRMS algorithm's reliable performance, highlighting its effectiveness in handling real data with promising outcomes.

Ultimately, the excellence of the MRMS method stems from the significant PG it delivers and its adaptability for implementation on programmable logic devices like FPGAs. The proposed MRMS algorithm excels due to its ability to achieve high PG while optimizing the use of implementation resources. In contrast to other analyzed techniques, achieving similar PG levels often requires greater resource utilization and higher system costs. The outcomes from the MRMS method can be considered as a clue for other researchers in the field to enhance the pulsed radar performance due to the implementation details and the practical results that are presented in this article.

Respecting the future work, we aim to use the MRMS technique on pulsed radars operating with other waveforms including classical and modern ones. Furthermore, we will examine how convex optimization methods are applied to lessen the Doppler intolerance of the used phase-coded waveforms for better enhancement. Another future work direction incorporates designing and utilizing an optimized denoising algorithm to mitigate the increase in the noise variance resulting from the addition process.

## REFERENCES

- [1] S. E. Abdelsamad et al., "Vision-based support for the detection and recognition of drones with small radar cross sections," *Electronics*, vol. 12, no. 10, p. 2235, May 2023, doi: [10.3390/electronics12102235](https://doi.org/10.3390/electronics12102235).
- [2] S. Rudys, P. Ragulis, A. Laučys, D. Bručas, R. Pomarnacki, and D. Plonis, "Investigation of UAV detection by different solid-state marine radars," *Electronics*, vol. 11, no. 16, p. 2502, Aug. 2022, doi: [10.3390/electronics11162502](https://doi.org/10.3390/electronics11162502).
- [3] W. Jiang, Z. Liu, Y. Wang, Y. Lin, Y. Li, and F. Bi, "Realizing small UAV targets recognition via multi-dimensional feature fusion of high-resolution radar," *Remote Sens.*, vol. 16, no. 15, p. 2710, Jul. 2024, doi: [10.3390/rs16152710](https://doi.org/10.3390/rs16152710).
- [4] Z. He, X. Chen, H. Zhang, L. Zhang, and C. Zhang, "Long-time integration for drone targets detection based on digital ubiquitous radar," in *Proc. IEEE 5th Int. Conf. Electron. Inf. Commun. Technol. (ICEICT)*, Aug. 2022, pp. 581–585, doi: [10.1109/ICEICT55736.2022.9909060](https://doi.org/10.1109/ICEICT55736.2022.9909060).
- [5] C. Cook, *Radar Signals: An Introduction to Theory and Application*. Amsterdam, The Netherlands: Elsevier, 2012.
- [6] N. Levanon and E. Mozeson, *Radar Signals*. Hoboken, NJ, USA: Wiley, 2004.
- [7] J. Gong, J. Yan, H. Hu, D. Kong, and D. Li, "Improved radar detection of small drones using Doppler signal-to-clutter ratio (DSCR) detector," *Drones*, vol. 7, no. 5, p. 316, May 2023, doi: [10.3390/drones7050316](https://doi.org/10.3390/drones7050316).
- [8] Y. Li, S. Yan, J. Gong, and F. Zeng, "SNR enhancement of back scattering signals for bistatic radar based on BeiDou GEO satellites," *Remote Sens.*, vol. 13, no. 7, p. 1254, Mar. 2021, doi: [10.3390/rs13071254](https://doi.org/10.3390/rs13071254).
- [9] S. Shen et al., "An improved coherent integration method for wideband radar based on two-dimensional frequency correction," *Electronics*, vol. 9, no. 5, p. 840, May 2020, doi: [10.3390/electronics9050840](https://doi.org/10.3390/electronics9050840).
- [10] M. Jawad, Y. Iqbal, and N. Sarwar, "Boosting SNR for passive detection of weak radar pulses under non-cooperative environment," in *Proc. Int. Bhurban Conf. Appl. Sci. Technol. (IBCAST)*, Jan. 2021, pp. 951–955, doi: [10.1109/IBCAST51254.2021.9393195](https://doi.org/10.1109/IBCAST51254.2021.9393195).
- [11] Y. Liu, G. Liao, S. Zhu, Z. Yang, Y. Chen, and X. Zhang, "Performance improvement in a coexistent radar and communications system," in *Proc. IEEE 11th Sensor Array Multichannel Signal Process. Workshop (SAM)*, Jun. 2020, pp. 1–5, doi: [10.1109/SAM48682.2020.9104303](https://doi.org/10.1109/SAM48682.2020.9104303).
- [12] A. Youssef, P. F. Driessen, F. Gebali, and B. Moa, "A novel framework for combining multiple radar waveforms using time compression overlap-add," *IEEE Trans. Signal Process.*, vol. 69, pp. 4371–4384, 2021, doi: [10.1109/TSP.2021.3094050](https://doi.org/10.1109/TSP.2021.3094050).
- [13] Z. Wu, G. Zhang, X. Wang, P. Sun, and Y. Lin, "An improved detecting algorithm of moving targets for airborne maritime surveillance radar," *Sensors*, vol. 25, no. 2, p. 560, Jan. 2025, doi: [10.3390/s25020560](https://doi.org/10.3390/s25020560).
- [14] B.-S. Kim, J. Lee, and S. Kim, "SNR and resolution improvement algorithm with the concatenation of multiple chirps for FMCW radar," *IEEE Antennas Wireless Propag. Lett.*, vol. 23, pp. 84–88, 2024, doi: [10.1109/lawp.2023.3317872](https://doi.org/10.1109/lawp.2023.3317872).
- [15] C. J. Nunn and G. E. Coxson, "Best-known autocorrelation peak sidelobe levels for binary codes of length 71 to 105," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 44, no. 1, pp. 392–395, Jan. 2008, doi: [10.1109/taes.2008.4517015](https://doi.org/10.1109/taes.2008.4517015).
- [16] G. Pawan, *Xilinx ChipScope Pro to Visualize FPGA Internal Signals*. LAP Lambert Academic Publishing, 2014.
- [17] IEEE Standard for Floating-Point Arithmetic, IEEE Standard 754-2019, 2019, pp. 1–84, doi: [10.1109/IEEESTD.2019.8766229](https://doi.org/10.1109/IEEESTD.2019.8766229).
- [18] Analog Devices. *ADALM-PLUTO*. Accessed: Feb. 13, 2025. [Online]. Available: <https://www.analog.com/en/resources/evaluation-hardware-and-software/evaluation-boards-kits/ADALM-PLUTO.html>
- [19] M. A. Richards, J. Scheer, and W. A. Holm, *Principles of Modern Radar*. Raleigh, NC, USA: Tes Dee Publishing, 2012.

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